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Risk Reduction of Concussion in Athletes: Do Neck Size or Neck Strength Make a Difference?

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Sean Silverman, DO: conceptualization, methodology, investigation, resources, writing, review/editing, visualization. **Meghan E. Vidt, PhD:** methodology, investigation, resources, review/editing, visualization, supervision. **Justin S. Hong, MD:** review/editing, supervision. **Lori M. Grafton, MD:** review/editing, supervision.

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Abstract

Medical provider's ability to detect, diagnose, and treat sport-related concussion (SRC) has greatly improved in recent years. Though more is known about the biomechanical forces involved in concussion, it is still uncertain whether there are preventative measures athletes can take to prevent a SRC from occurring. The objective of this review was to determine if either neck size or neck strength is related to a decreased risk of sustaining a SRC. A literature review was conducted on Google Scholar and Ovid MEDLINE for pertinent articles. Findings indicate that isometric neck strength, but not neck size, has been shown to be a predictor for SRC prevention. Formal neck strengthening programs are feasible and lead to decreased SRC risk. Additionally, there may be greater opportunity to increase neck strength in amateur athletes compared to professional. In conclusion, cervical strengthening programs have been shown to be feasible and beneficial for athletes to decrease their risk of sustaining a SRC, though the optimal duration, intensity, and frequency has yet to be determined.

Keywords: brain concussion, concussion, primary prevention, neck strengthening, cervical strengthening

List of Abbreviations

mTBI, Mild Traumatic Brain Injury

SRC, Sport-related concussion

AE, athletic exposure

U18, under age 18

FIFA 11 +, Federation Internationale de Football Association's 11+ Injury Prevention Program

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Introduction

Though there is no universally accepted definition of mild traumatic brain injury (mTBI), it is generally defined as an acute brain injury resulting from mechanical energy to the head from external physical forces[1, 2]. Diagnostic criteria include loss of consciousness less than 30 minutes, post traumatic amnesia less than 24 hours, a Glascow Coma Scale score of 13 to 15[2], as well as mental status and behavioral changes, and/or neurological signs[3]. Though the clinical presentation of mTBI can vary, often the terms mTBI and concussion are used interchangeably[3]. Of particular interest in the athletic population is sport-related concussion (SRC). SRC is defined as brain injury resulting in the disturbance of typical brain functioning and involves a complex pathophysiological process[4-6]. SRC incidence is estimated to be around 3.8 million injuries per year in the United States[7]. At the high-school level, the incidence rate of SRCs is highest in boys' football (10.40 per 10,000 athletic exposures [AEs]), followed by girls' soccer (8.19 per 10,000 AEs) and boys' ice hockey (7.69 per 10,000 AEs)[8]. SRCs can also occur at all levels of low contact sports (e.g. tennis, swimming, track and field). The true incidence of SRC is likely higher than documented, as many athletes may not recognize or report their SRC[9-11]. Notably, the Center for Disease Control and Prevention claims that five out of ten concussions go undiagnosed or undetected[12].

The biomechanical forces involved in concussion include direct impact to the head, or an impact to another part of the body with a mechanical force transmitted to the head[13]. These forces can be linear or rotational[14]. In addition to increased acceleration, the direction of acceleration is also important, as head impacts in the real world occur in all planes of motion. There is evidence to suggest that coronal plane acceleration (lateral flexion) may be more

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injurious than sagittal plane acceleration (flexion and extension)[15]. Ultimately, a better understanding of the biomechanical forces that cause concussion may lead to more targeted prevention approaches.

Although the medical community's ability to detect, diagnose, and treat SRC has greatly improved, the ability to prevent SRCs from occurring and decrease the severity of clinical outcomes post-injury is limited[16]. If the acceleration of the brain during an impact can be decreased[14], it has been theorized that this could prevent a concussion from occurring. This is an important concept as the greatest opportunity to impact overall injury burden often lies in injury prevention[14]. However, the literature is sparse in regard to concussion prevention in athletes. Patricios et al. has recently published a revised consensus statement on concussion in sport. While there were five unanimously-supported recommendations for concussion prevention (i: Mouthguard use should be supported in child and adolescent ice hockey, ii: Policy disallowing body checking should be supported for all children and most levels of adolescent ice hockey, iii: Strategies limiting contact practice in American football should inform related policies and recommendations for all levels, iv: neuromuscular training warm-up programmes are recommended, based on research in rugby, and more research is needed for female athletes and in other team sports specifically targeting exercise components aimed to reduce concussion rates, v: Policy supporting optimal concussion management strategies to reduce recurrent concussion rates is recommended)[17], none of the recommendations included cervical strength. Four of the recommendations were specific to a singular sport and the fifth was a general comment about policy change[17]. Thus, there are no agreed upon guidelines or formal recommendations in regard to cervical strength or cervical strengthening programs and athlete concussion reduction.

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Neck size and neck strength have emerged as modifiable characteristics associated with concussion risk[12, 14]. Neck size is an easily measured variable given that neck circumference can be obtained with simple and accessible tools, like a measuring tape. Isometric neck strength, though a more difficult variable to test due to the need for equipment, such as a dynamometer, may be more clinically useful because it is a more functional characteristic as opposed to neck size alone, which is static measurement. There also may be certain athlete populations that could benefit from a formal cervical strengthening program, since they have weaker necks[18, 19], including amateur athletes and female athletes. Due to the paucity of research and data on this topic, the purpose of this narrative review is to examine all the literature available regarding neck size and/or neck strength and its relation to SRC prevention and if a formal cervical strength training program decreases SRCs.

Methods

A search of Ovid MEDLINE database was performed with the key terms outlined in box A, resulting in 13 articles. Google Scholar was utilized to search for additional articles with key terms outlined in box B, resulting in 552 articles. An additional filter to exclude articles prior to the year 2000 was used, which resulted in 485 articles. As there is a paucity of research in this area spread across a variety of sports and athlete populations, the inclusion criteria was intentionally kept broad to ensure we captured the literature available on the topic. All articles relating to sports related concussion and neck size and/or neck strength were included, regardless of the sport, gender of the athletes, or age of the athletes. The primary author reviewed the remaining 485 articles and included all 22 articles related to sports related concussion. Two articles were later removed during the article revision process as they were outside the scope of

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this manuscript. Combining the 20 articles from Google Scholar and 13 articles from Ovid MEDLINE resulted in a total of 33 articles included for this literature review (Figure 1).

Discussion

Neck Size

Neck size, typically measured as neck circumference, was theorized to be a protective factor for concussion, as a larger neck could mechanically resist more force and prevent a concussion from occurring. Neck circumference has been proposed as a simple proxy measure of other features of the cervical spine, such as neck strength[20]. However, the literature does not support this theory. Neck circumference has not been shown to be related to SRC history or subsequent SRC. Since neck size is not related to SRC, additional studies examined proportional neck circumference (neck circumference in proportion to body mass)[20]. Similarly, there was no relationship to proportional neck circumference and reduction of SRC. Males had a significantly larger neck circumference and proportional neck circumference compared to females, but when controlling for sex, there was no association between neck circumference and SRC[20]. Eckner et al. found possible evidence that increased neck size may correlate with reduced concussion risk in high-school aged male athletes involved in contact or collision sports[21]. Their results showed that neck girth, quantified by measuring the circumference of the neck at the level of the thyroid cartilage, was associated with decreased SRC but only with an increase in isometric neck strength. Collectively, larger neck girth was only associated with decreased SRC if there was also a concomitant increase in isometric neck strength. There were also findings that an increase in neck strength without a change in neck girth also led to decreased concussion risk, but the opposite was not true[21]. One must be cautious when simply attributing neck circumference to

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decreased risk of SRC after a cervical strengthening program, as it appears the more pertinent variable is neck strength. Eckner et al. has suggested a theory that initial strengthening may not lead to significant changes in neck size, but significant changes in strength. This theory is consistent with the effect of weight lifting regimes on extremity muscles, where initial strength gains are from neural adaptation rather than muscle hypertrophy[21].

Isometric Neck Strength

There is markedly more evidence showing that isometric neck strength is a protective factor from SRC. Cervical musculature is primarily responsible for maintaining posture and stabilizing the head at rest[22]. Cervical muscle contraction is theorized to more rigidly couple the head to the torso, and therefore decrease the acceleration experienced by the brain[14]. Neck stiffness, via the cervical musculature, is the dynamic ability of the neck to reduce translational head displacement, velocity, and acceleration[15]. Individuals who have stronger necks can more effectively distribute potentially injurious forces from impacts compared to those with weaker necks[23]. Greater neck strength has been shown to decrease acceleration of the head during rugby contact events[24]. In particular, the sternocleidomastoid muscle has been shown to be the single muscle making the greatest contribution to force generation across all planes of cervical spine motion[21]. Head position at impact can also result in greater forces to the brain. Forward head posture is defined as the external auditory meatus being positioned anterior to the glenohumeral joint. This particular head position will increase activation of the sternocleidomastoid and upper trapezius and subsequently inhibit the deep muscles responsible for segmental stability and neck proprioception. This may result in increased head acceleration during impact due to the muscle imbalances noted in this posture[16].

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Not only is strength important, but the symmetry (i.e. balanced strength between neck flexor and extensor muscle groups) of that strength must be considered. Symmetrical strength in neck flexors and extensors reduces head acceleration during low-velocity soccer heading in experienced collegiate players[18]. Muscle stiffening can allow both the head and torso to absorb kinetic energy, as opposed to just the head when muscles are not stiffened. This minimizes acceleration to the head by increasing the relative mass of the head-neck segment since the head is better coupled to the torso[18, 25]. Dezman et al. found that symmetry imbalance in neck musculature (flexors versus extensors) in the sagittal plane led to higher head accelerations[18, 23]. Another way to decrease the acceleration of the head is to extend the time in contact with the ball (i.e. while heading a soccer ball)[18]. Acceleration is a result of the speed of an impact per unit of time; therefore, extending the impact over a longer time period will decrease the acceleration of the head[16]. Contracted neck muscles are then better able to absorb and dissipate the applied force on the head down through the neck to the body, thereby minimizing the resultant acceleration of the head. Additionally, peak linear acceleration was shown to decrease in players who pre-tensed their neck muscles compared with players who did not[25]. Collectively, this means that those soccer players who were able to contract their neck muscles prior to heading a soccer ball had lower acceleration forces to the head, and thus would be at lower risk of a concussive force.

A correlation has been shown between neck strength and head impact kinematics, where individuals with higher neck strength measurements had lower head accelerations upon impact[26]. Imbalances in neck flexor and neck extensor strength ratios in female collegiate soccer athletes have been correlated with greater angular acceleration when heading the ball. Less overall neck strength has been significantly correlated with higher concussion rates in basketball, lacrosse,

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and soccer[27]. Weaker mean overall neck strength was significantly associated with concussion, and for every pound of neck strength that athletes gain, their odds of a concussion decreased by five percent[26]. There is growing evidence to suggest that neck strength is protective against SRC, although methods to measure neck strength can be cost prohibitive. For example, in Attwood et al.'s study the hand held dynamometer used to measure neck strength costs approximately $1000 (US dollars). The associated cost is potentially too expensive for a majority of athletes and/or teams, presenting a barrier to access[24]. In the future, it would be beneficial to develop an accessible way to measure neck strength, so these measures can be obtained and used across a larger patient population. In addition, if there were standardized values of "at risk" neck strength, these athletes could be more readily identified. Therefore, we must look toward cervical strengthening programs to determine if they are feasible to carry out and effective at decreasing SRC. Currently, there is no consensus of whether one type of strengthening program is preferred over another (i.e. isometric versus eccentric) or if a strengthening program leads to a decrease in SRC.

Formal Isometric Neck Strengthening Programs

There has been a notable increase in the number of studies assessing whether isometric cervical strengthening programs can be successfully implemented in athletes to increase their neck musculature in addition to a decrease in SRC. Waring et al. discovered that a six-week cervical neck strengthening program only improved muscle strength in the anterior and anterolateral (right and left) directions[26]. Toninato et al. concluded that greater neck strength was associated with a decreased kinematic response of the head to controlled impulsive loading[28]. Fisher et al. had an eight-week cervical strengthening program where exercises were performed

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twice per week[19]. They concluded that there is potential to significantly improve neck strength using a simple, low volume resistance training protocol[19]. However, these formal neck strengthening programs could potentially be an unrealistic regime, as they are commonly burdensome, require specialist equipment, or a large time commitment, all of which can be barriers to implementation. Neck muscle function can be altered with targeted resistance training, but the time and equipment demands are barriers to their implementation[24]. To combat this issue, Attwood et al. performed a study with a under age 18 (U18) male soccer team[24]. Athletes performed isometric neck exercises three times per week for eight weeks. These included maximal contraction of the neck in each direction (flexion, extension, left/right lateral flexion) with resistance provided by their own hands. Total time for exercise was only three minutes and did not require any special equipment. The intervention group's total neck strength demonstrated a significant 24% increase over that of the control group. The two major conclusions from the study were: 1) performing a neck exercise program independently, or as part of a whole-body program, could contribute to lower sports-related head and neck injuries; and 2) near-maximal to maximal loads may be required to induce meaningful changes in neck strength when considering isometric or isotonic neck exercises[24]. An additional study by Peek at al. reflected the ease with which a neck strengthening program can be instituted for soccer players[25]. The rationale for this study was that players with reduced neck strength often demonstrate an inability to generate the necessary stiffness of the neck column to control the head's response during purposeful heading. Male and female adolescent soccer players who completed a five-week neuromuscular neck exercise program integrated into part two of the Federation Internationale de Football Association's 11+ Injury Prevention Program (FIFA 11 +) demonstrated an increase in isometric neck strength and a decrease in head impact magnitude during purposeful heading. Despite

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adding less than two minutes to part two of the FIFA 11 +, boys and girls in the intervention group were able to demonstrate large increases in neck strength in only five weeks[25]. Most notably, strength increased in neck extensors, with a reported effect size of 0.22 and 95% confidence interval of (0.08–0.34)[25]. The conclusion of this study was that players who completed neck exercises demonstrated an increase in isometric neck strength and a decrease in head impact magnitude during heading. Importantly, these exercises were easily incorporated into the athletes' usual training. Eckner et al. found that neck strength increased in the intervention group compared to the control group[21]. They also concluded that the risk of SRC was significantly reduced with a neck strengthening exercise program. They were also able to conclude that a five percent decrease in odds of concussion were reported for every one-pound increase in neck strength[21]. Despite the evidence suggesting the benefits of reduction of SRC with a simple, efficient, self-performed cervical strengthening program, it does not appear to be a popular technique implemented by sports teams or athletes. Jeffries et al. discovered that although nearly 70% of collegiate women's soccer athletic trainers believed that cervical strengthening programs would aid in concussion prevention, only 17% currently used such programs[27]. Future research could be utilized to determine the optimal exercises, frequency, duration, and intensity of a cervical strengthening program so that it can be implemented into an athlete's training to reduce concussion risk.

Non-Professional Athletes' Cervical Strength

In regard to concussion prevention, it is important to consider the differences between professional and amateur athletes because there is conflicting evidence about whether all athletes would benefit from a cervical strengthening exercise program. There are inherent differences

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between professional athletes compared to amateur athletes (size, strength, speed, skill, etc). For instance, Attwood et al. concluded that youth rugby players have significantly lower neck strength compared with adult rugby players, which may predispose these younger players to injury, particularly when transitioning to the adult game[24]. Additionally, there are some physical attributes that predispose children to concussion. For example, Eckner et al. remarked that children may be at an even greater disadvantage to sustain a concussion than adults because of their greater head mass to body mass ratio[15]. In a similar study performed by Eckner et al., it was discovered that a less highly trained youth population training in all three planes of motion might have greater potential for improvement than the elite collegiate athletes[21]. They found that youth athletes were able to decrease the rotational accelerations to their head with a cervical strengthening program[21]. Although no explanation or causation for this finding was provided, Zasler et al.[4] remarks that the incidence rate of SRCs is lower for football at the collegiate level (4.00 per 10,000 AEs) than the high school level, despite higher injury overall injury rates (e.g., orthopedic, musculoskeletal) in collegiate players. Collectively, this highlights how younger athletes tend to be more at risk for SRC due to certain non-modifiable risk factors. Specifically, younger athletes may have a greater ability to reduce their concussion risk compared to professional athletes, since they have relatively weaker necks and, thus, a larger potential to increase their neck strength.

Rotto et al. studied high school football players and determined that a seven-week cervical strengthening regime prior to the football season led to increased neck strength and reduced concussions[29]. The seven-week program was incorporated into the team's pre-season strength and conditioning workouts. Ultimately, 54 athletes participated in the study; while the

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team suffered a total of four concussions throughout the season, two of those concussions were experienced by athletes who did not participate in the strength program. Compared to 12 concussions in the year prior and seven in the season before that, there was a significant reduction in SRC following the strengthening program[29]. This study suggests that a cervical strengthening program can have a direct clinical effect on reducing SRC in athletes, although larger and more robust trials are needed to strengthen this claim. Hamlin et al. found that a simple 6-week neck strengthening program improves maximal voluntary isometric contraction strength in male amateur rugby players[30]. The evidence has shown that non-professional athletes tend to have greater risk reduction from concussion when a cervical strength program is employed.

There is currently a paucity of data describing the relationship between neck strength and SRC risk, with current research only delineating between professional and non-professional athletes. More research is needed to fully understand the role of neck strength and SRC in athletes of all levels, including youth, high school, collegiate, and professional. It has been suggested that a neck strength assessment could be used as a preparticipation screening for athletes to identify those who are most at risk for SRC so a strengthening regime can be proactively implemented into the athlete's training regimen[18].

Male versus Female Cervical Strength

Cervical strengthening programs may result in different outcomes based on the sex of the athlete. Even the updated consensus statement on concussion admits that more research is needed for female athletes in regard to concussion prevention[17]. Prevention strategies may need to be

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tailored to account for inherent physical characteristics and morphological differences in neck traits between males and females. For example, the finite element neck model by Braun et al.[31] may provide some insight on these differences. Female vertebrae from C3-7 are significantly smaller than male vertebrae in the anterior-posterior dimension. Males also have more linear and areal dimensions of the cervical spine compared to females[22]. This could indicate males have a more stable spinal column that could resist inertial loads. Females also have greater ligamentous laxity and smaller neck muscles than males[22]. Ligamentous laxity predisposes a joint to injury as it is inherently more unstable, and relies more on dynamic stabilizers (i.e. muscles) to prevent injury. A study performed by Carmo et al. examining whiplash injury in motor vehicle collision showed that females had higher risk of concussion[22]. Although not sports related, this study suggests that the external forces combined with the female anatomy could lead to differing exposure to head injury than males. More work is needed to translate the findings of Carmo et al. to the study of female anatomical characteristics and their role in SRC.

Gallo et al. has compiled several differences in cervical anatomy and function between the sexes[32]. Males had significantly greater peak force and greater time to peak force versus females for sternocleidomastoid muscles. Males have greater overall cervical muscle strength compared to females, while females have greater head-neck segment acceleration and displacement than males when exposed to an external force to the head. They also concluded that there are differences between the sexes in their cervical muscle firing patterns and rates[32]. Future cervical strengthening programs may need to consider these factors to tailor programs to males and females.

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The preceding evidence highlights findings in literature to suggest that females experience SRC more often than males, and they may have a larger reduction in risk if a cervical strengthening program is implemented. Jeffries et al. concluded that female soccer players have a concussion incidence rate 2.4 times greater than that of male soccer players, where the majority of concussions are caused from contact with the ball as opposed to contact with another player[27]. The original data came from a case-control study performed by Fuller et al. that included 20 FIFA tournaments over 6 years resulting in 19,802 player hours of match exposure, which included 15,115 hours for men and 4,687 hours for women[33]. They also found that those athletes with weaker neck strength were more likely to suffer concussion, and that females had weaker necks compared to males[27]. Peek et al. found similar results whereby female collegiate soccer players exhibited significantly lower neck strength and greater head accelerations during purposeful heading than males[25]. In the future, literature will need to address these differences between the sexes. In order to prevent SRC there may need to be specific exercises tailored toward males and females. For example, Daly et al. concluded there must be a more comprehensive understanding of the etiology of concussion in females[13] because not only are females more at risk for SRC, but they report greater symptom severity and longer recovery times than males. Striefer et al. had similar findings, showing that females experienced more severe symptoms and longer recovery patterns post-SRC than males[16]. The fact that females suffer concussion at a greater rate and have more significant symptoms with prolonged recovery times compared to males highlight the importance of identifying and implementing preventative measures to offset SRC risk. Although the literature is currently sparse, there is emerging evidence that neck strengthening exercises can help offset risk of experiencing SRC. Prior work shows that females achieved greater relative strength increases than males following a

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strengthening regimen, suggesting that the degree of disparity can be reduced[19]. It has been hypothesized that females have greater neuromuscular activation to account for their lower neck strength, which could be one possible mechanism female athletes may rely on for concussion prevention[23]. Ultimately, there appears to be a disparity between males and females in regard to SRC occurrence. This may prove clinically important in the future, as treatment and preventative measures may also differ between the sexes.

Conclusion and Limitations

Concussion management has centered around treatment after the injury has occurred, but it is well known in medicine that injury prevention is the ultimate goal. Primary prevention in regard to SRC is in its infancy, but has some emerging topics. At the forefront is neck or cervical characteristics, which may serve as protective indicators. However, the evidence to support neck circumference or size as a protective factor is still lacking. Though an easily measurable trait with a possibility for a quick screening tool, increased neck size has not yet shown any relationship to concussion prevention. On the contrary, neck strength does have supportive evidence to be a protective factor against concussion. More importantly, it has been shown that a simple neck isometric strengthening program is able to increase neck strength and decrease concussion risk. However, caution must be exercised as strengthening programs appear to be most helpful in certain populations, like females achieving greater benefit than males, and amateurs compared to professionals. Similarly, due to the functional and anatomical differences between the sexes, different exercise programs tailored to sex are needed. It is also unclear if a cervical strengthening program would benefit athletes in all sports, and the duration, intensity, and frequency of a cervical strengthening program has yet to be determined.

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Figure Captions

Figure 1: Article search strategy, including the key terms that were used to search (A) the Ovid MEDLINE database, and (B) the Google Scholar database. After applying the search criteria and reviewing the articles consistent with the inclusion criteria, a total of 33 articles were evaluated in the current review.

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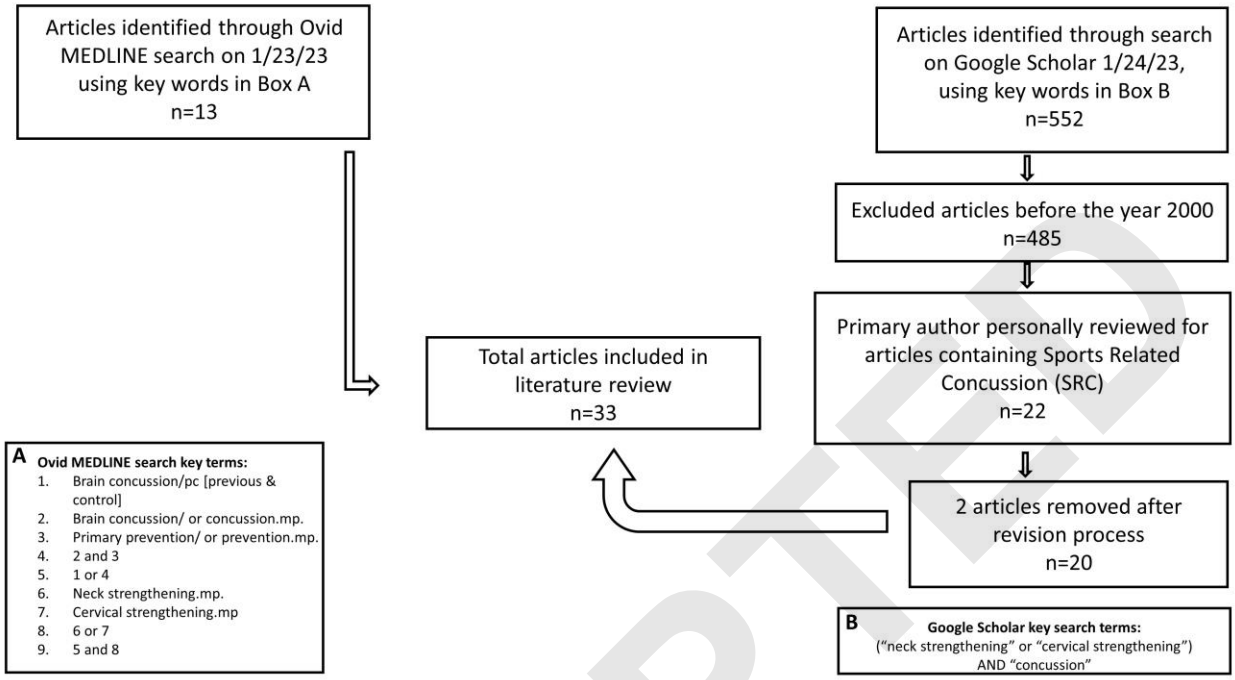
Figure 1



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